

Multiple Choice Questions (MCQ)

IIoT IN 2025 Seminar: All Sections (1-4)

Assimilation Aid for Protocol and Data Challenges

Novembre 2025

Multiple Choice Questions (MCQ) - Section 1: From the Smart Factory to Big Data

Please refer to the indicated slide numbers for verification.

1. What best characterizes a Smart Factory? [Slide 8]

- A. A production facility using only robots for assembly.
- B. A highly digitized facility that integrates **OT (factory floor control)** with **IT (Cloud/enterprise analytics)**.
- C. A factory that replaces all old systems by modern protocols like OPC UA.
- D. A factory that collects data only for corrective maintenance.

2. In the context of IIoT, what is the general process that transforms a Raw Signal into an Action (Insight)? [Slide 3]

- A. Capture → Transformation → Recording
- B. Capture → **Transformation (Analysis/Calculation)** → **Insight (Action)**.
- C. Capture → Transmission → Visualization
- D. Command → Execution → Report

3. In the Predictive Maintenance example (Vibration), which step corresponds to the Transformation (Analysis)? [Slide 3]

- A. The acceleration sensor records measurements every millisecond.
- B. The system sends an alert: the motor bearing is likely to fail within 7 days.
- C. **Frequency Analysis (FFT)** is performed and compared to the baseline (normal state).
- D. The spare part order is placed.

4. According to Slide 6, the "Transformation" step is essential at the Host Layer (IT/Cloud) to create an Object Model with ID. What is the main goal of this step? [Slide 6, 12]

- A. Calculate advanced statistics like Frequency Analysis (FFT) of vibrations.
- B. Ensure real-time safety and reliability (low latency) of production.
- C. Replace old protocols like Modbus with modern protocols like HTTP.
- D. **Assign a unique ID, a unit of measure (°C, kWh) and an IT-readable context to a raw value (e.g., Register 40001).**

5. What is one of the main challenges of Big Data illustrated by the "V"s in the presentation? [Slide 9]

- A. Complexity and Compatibility.
- B. Cost and Conservation.
- C. **Volume** and **Velocity (Speed)** of data.
- D. Authenticity and Accessibility.

6. What is the main priority of the OT (Operational Technology) domain compared to the IT domain? [Slide 12]

- A. Data Confidentiality and Scalability.
- B. Security (cybersécurité) and Encryption.
- C. **Safety** and **Real-Time Reliability (Low Latency)**.
- D. Standard TCP/IP networking.

7. How does the Data Context differ between OT and IT, thus creating a gap? [Slide 12]

- A. OT uses only analog data, IT uses digital data.
- B. OT manages **raw values** (e.g., Register 40001 = 85.5) and IT uses an **Object Model** (e.g., Motor Temperature = 85.5°C).
- C. OT uses Modbus, IT uses OPC UA.
- D. OT works in Hexadecimal, IT works in Decimal.

8. The OPC UA protocol is crucial because it is “born from IT principles”. What does this latter point mean? [Slide 14]

- A. It is simple and deterministic like Modbus.
- B. It uses the **TCP/IP** protocol and integrates **encryption (TLS)** to meet the **security** requirements of the IT domain.
- C. It replaces all old Modbus and Profinet protocols.
- D. It works only on Layer 1 (Physical) of the OSI model.

Multiple Choice Questions (MCQ) - Section 2: The Enduring Utility of Modbus

Please refer to the indicated slide numbers for verification.

9. What is the main reason for the persistence of the Modbus protocol in OT Controllers despite its lack of native security? [Slide 17, 33]

- A. It is the only protocol that supports TLS encryption for local communication.
- B. It is designed on a Client/Server architecture, ideal for modern networks.
- C. Its characteristics of **Simplicity (Low Complexity)**, **Ubiquity**, and **Predictability (Determinism)** make it essential for real-time control.
- D. It is faster on Ethernet than the OPC UA protocol.

10. What is the historical communication model on which the Modbus protocol is based for data exchanges at the industrial device level? [Slide 15]

- A. Publish/Subscribe (Pub/Sub)
- B. **Master/Slave**
- C. Client/Server
- D. Peer-to-Peer

11. What is the main role of the Modbus protocol in the industrial architecture (OT)? [Slide 15]

- A. Ensure data modeling for the Cloud (IT).
- B. Provide end-to-end encrypted security.
- C. Serve as a **fundamental protocol for low-level communication** within industrial devices (PLCs, meters, sensors).
- D. Replace the TCP/IP architecture with serial communication (RS-485).

12. One of the Modbus variants is Modbus Plus (Modbus+). What major advantage was this variant designed to guarantee in OT environments? [Slide 16]

- A. Native support for encryption (TLS).
- B. Transmission of Modbus data over Wi-Fi networks.
- C. **Ensure guaranteed network access via a token passing mechanism** for deterministic control loops.
- D. Support for millions of devices (Big Data scalability).

13. When Modbus is used over Ethernet (Modbus on TCP/IP), what essential component is added to the standard Modbus message to support the TCP/IP network? [Slide 16, 25, 27]

- A. The Security Layer (TLS).
- B. The OPC UA Layer.
- C. **The MBAP Header (Modbus Application Protocol Header)**.
- D. A 32-bit CRC (Cyclic Redundancy Check).

Multiple Choice Questions (MCQ) - Section 3: Let's Build a Scenario

Please refer to the indicated slide numbers for verification.

14. In the context of the scenario (Diapo 19), what is the primary function of the Data Gateway (or Protocol Converter) located between the PLC/Sensor and the Cloud/IT infrastructure?

- A. To manage the entire factory production schedule (MES).
- B. To directly run real-time control loops (OT).
- C. To translate the low-level OT protocol (e.g., Modbus) into an IT-friendly format (e.g., OPC UA or MQTT) and add context (Object Model).
- D. To physically separate the Ethernet cables from the serial cables.

15. The scenario (Diapo 19) illustrates the presence of major IT/Cloud platforms (e.g., Siemens MindSphere, AWS/Azure). In this architecture, what role does the Modbus protocol play? [Diapo 21]

- A. It serves as the primary protocol for all data transfer to the Cloud.
- B. It provides the secure link between the Gateway and the Cloud.
- C. It is strictly an **Internal Protocol** used only for basic, low-level data extraction within the factory (between the Gateway and the devices).
- D. It models the data using Object-Oriented Programming (OOP) concepts.

16. When Modbus TCP/IP is used, the traditional Master/Slave model is conceptually replaced by which modern communication model? [Diapo 21, 22]

- A. Peer-to-Peer
- B. **Client/Server**
- C. Publish/Subscribe
- D. Broadcast

17. In a Modbus TCP frame, the MBAP Header includes the 'Transaction Identifier' and the 'Unit Identifier'. What is the purpose of the Transaction Identifier?

- A. It specifies the Function Code (read or write).
- B. It determines the IP address of the Slave device.
- C. It allows the Modbus Server to **match the Request frame with its corresponding Response frame**.
- D. It indicates the physical device address on the RS-485 bus.

18. In the context of a Modbus TCP request, what does the Function Code contained in the PDU (Protocol Data Unit) primarily determine? [Diapo 25, 27]

- A. The security level (TLS or not).
- B. The speed of the connection (baud rate).
- C. **The type of action to perform (e.g., Read Holding Registers or Write Single Coil).**
- D. The number of connected devices (Slaves).

Multiple Choice Questions (MCQ) - Section 4: Future IIoT Adaptation and Vertical Applications

Please refer to the indicated slide numbers for verification.

19. What is the primary additional challenge when adapting IIoT protocols for the Internet of Medical Things (IoMT) compared to standard industrial applications? [Diapo 32, 33]

- A. Handling extremely high data volume (Big Data).
- B. The strict need for low-latency control loops for machine safety.
- C. **Ensuring high levels of patient data privacy and security (e.g., HIPAA compliance).**
- D. The difficulty in connecting devices via Wi-Fi.

20. Vehicular Ad-Hoc Networks (VANET), as an IIoT vertical, primarily rely on which type of communication model? [Diapo 32, 33]

- A. A fixed Master/Slave hierarchy.
- B. Direct, centralized communication with a single cloud server.
- C. **Decentralized, dynamic communication between vehicles (V2V) and infrastructure (V2I).**
- D. Long-range satellite communication only.

21. In specialized IIoT vertical applications like IoMT and VANET, which widely adopted IT protocol is crucial for secure, high-level data transport and object modeling, bridging the gap from device data? [Diapo 14, 32]

- A. Modbus.
- B. RS-485.
- C. **OPC UA or MQTT (for lightweight, context-rich data exchange).**
- D. Profibus.

22. When deploying an IoMT system for remote patient monitoring, the OT domain's priority of 'Safety' translates into which concern? [Diapo 12, 32]

- A. Ensuring the system is highly encrypted to prevent data theft.
- B. Making sure the system is scalable to millions of users.
- C. **Ensuring continuous, real-time data flow (low latency) from life-critical sensors to prevent immediate harm.**
- D. Using only Modbus TCP/IP for communication.

23. Why is 'low latency' a critical requirement for safety applications (e.g., collision avoidance) within Vehicular Ad-Hoc Networks (VANET)? [Diapo 12, 33]

- A. To minimize data storage costs in the Cloud.
- B. To allow the use of older, simpler Modbus protocols.
- C. **Because the time delay in receiving and processing traffic or hazard warnings must be minimal to enable timely vehicle reaction.**
- D. To ensure high visual quality of the in-car navigation screen.

Supplementary Resources and References

1. **Original Presentation :** IIoT IN 2025 : Mastering Massive Data Challenges with Industrial Protocols. By Khaoula Ben Ali, Ph.D. (ENIT, ENSEEIHT), Researcher at Informatic Laboratory of Industrial Systems, (LISI) University of Carthage.
2. **Modbus Video Example :** An illustration on Modbus frames (ADU and PDU) is available on YouTube : <https://www.youtube.com/watch?v=JBGaInI-TG4> [Slide 24].
3. **Advanced Topics :** The basic IIoT architecture presented in the seminar is applicable to emerging fields such as the Internet of Medical Things (IoMT) and Vehicular Ad-Hoc Networks (VANET) [Slide 33].